

NICE ASSISTANT - Project Summary

Generated: March 7, 2026 | Project Root: F:/Hackathon Project

1) Project Purpose

Nice Assistant is an Android-first, offline-default assistant focused on privacy, local intelligence, and practical on-device actions. It performs intent routing, file indexing/search, OCR, wake-word/STT/TTS, automation, and contextual response generation without sending user data to cloud services by default.

2) Core Capabilities Implemented

- Offline-first policy with explicit session opt-in for internet features when needed.
- Deterministic intent parser with direct task execution (alarms, timers, notes, app launch, file open/search).
- Content-first retrieval over local files: BM25 candidate selection + embedding rerank + answer quality guard.
- Robust file extraction pipeline for TXT/PDF/IMG/DOCX with Android read fallback (readFile -> getUri -> fetch).
- Knowledge graph memory with citations and confidence-aware fallback responses.
- Mic arbitration + wake-word / STT lifecycle hardening with fallback pathways.
- On-device automation rules: trigger-action execution fully offline.
- Performance governor for low-end phones (RAM/CPU + battery/data-saver aware tuning).
- Encrypted personalization memory with explicit user consent control in settings.
- Multimodal capture flow for image/screenshot OCR and follow-up Q/A.

3) Architecture Snapshot

- UI Layer: Vanilla HTML/CSS/JS + Vite
- Runtime: Capacitor Android bridge for native integration
- Search/RAG: Local index + BM25 + embedding worker + confidence verifier
- Storage: IndexedDB + localStorage (encrypted where applicable)
- ML/OCR: ONNX runtime web, transformers-based embeddings, Tesseract OCR, PDF.js, Mammoth DOCX
- Actions: alarms/timer/media/system controls + file open routes

4) Security, Privacy, and Reliability Hardening

- Offline default preserved; online features require explicit session enablement.
- Endpoint allowlisting and policy-gated network calls.
- No simulated success paths in production responses.
- Safe action planner blocks unsafe/non-deterministic plan steps.
- Regression suites added for policy behavior, retrieval hardening, worker lifecycle, and stability forecasts.

5) Validation and Release Status

Area	Status
Parser/Math/Intent tests	PASS
Policy runtime tests	PASS
Search hardening tests	PASS
Embedding worker tests	PASS

Offline-brain/next-feature/stability tests	PASS
Production web build	PASS
Signed Android release build	PASS

6) Current APK Outputs

Latest stable signed APK: F:/Hackathon Project/release/Nice-v1-signed-20260307-stable.apk
Package: com.example.hackathon | Target SDK: 36 | Build channel: direct sideload

7) Operational Notes

For full file coverage and floating bubble behavior on Android 13+, users must allow special permissions: Manage All Files Access and Display Over Other Apps. Standard runtime permissions (e.g., microphone) were verified in deployment testing.

Conclusion: The project is in a deployable stable state with offline-first intelligence, hardened retrieval, privacy controls, and signed APK output.